

Documentation

1. Introduction

The **Autonomous Campus Guide** is a robotic assistant that combines speech recognition, **Large Language Models (LLMs)**, **navigation AI**, and contextual data sources (**maps, FAQs, events**) to support students and visitors with navigation and campus information.

2. Actors & Roles

- **Primary Actor:** Student, newcomer, visitor
- **System:** Robot assistant (speech recognition, LLM, navigation AI)
- **Supporting Systems:**
 - **Context Broker (FIWARE NGSI-LD)** → indoor maps, building coordinates
 - **FAQ Knowledge Base** → structured campus data (cafeteria, events, etc.)
 - **MCP Servers** → modular AI services (LLM, NavAI, VLM)
 - **n8n Automation** → orchestrates workflows (update FAQs, fetch events)

3. System Workflow (Technology Translation)

1. **Voice Input**
 - a. Hardware: robot microphone
 - b. Software: Speech-to-Text (Whisper, Vosk, Google Speech API)
2. **LLM Processing**
 - a. Local (Jetson Nano) or cloud MCP server
 - b. Detects intent: navigation vs FAQ vs small talk
 - c. Queries Context Broker or FAQ DB
3. **Workflow Orchestration**
 - a. FIWARE ROS Agent bridges robot ↔ Context Broker
 - b. MCP Servers run modular services (LLM, NavAI, VLM)
 - c. n8n handles external integrations (events, news feeds)

4. Response Generation

- a. Navigation → NavAI → converts path into steps
- b. FAQ → DB/n8n returns structured info
- c. Small Talk → LLM generates natural dialogue

5. Voice Output

- a. TTS (Coqui TTS, eSpeak, Amazon Polly)
- b. Played on robot speakers + optional LCD display

4. Role of VLM (Vision-Language Models)

- Identify landmarks visually: “Cafeteria is the building with red doors”
- Confirm destination: “Here’s the robotics lab entrance”
- Aid localization: match live camera view with map

5. Deployment Modes

- **Fixed Robot (Info Desk)** → stays at entrance, provides voice guidance
- **Mobile Escort (future)** → physically guides students through halls

6. Value

- **Practical:** Easy navigation and info access
- **Engaging:** Conversational and human-like interaction
- **Modular:** MCP allows independent upgrades of LLM, NavAI, VLM
- **Scalable:** FIWARE + n8n enable event/news integration